

Image Classification

Linear Classifier

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Lecture Outline

- Image Classification
- Challenges for Image Classification
- Image Classification Extended to other Tasks
- Logistic Regression for Image Classification
- Neural Networks for Image Classification

Image Classification

- A core task in computer vision.

$$I_{RGB} \rightarrow \mathbb{R}^{H \times W \times 3}$$

$$\text{Dataset } \mathcal{D} = \left\{ \underbrace{(x^{(i)}, y^{(i)})}_{\substack{\text{image} \quad \text{ground truth} \\ \text{class label}}} \right\}_{i=1}^N$$

Input: image



This image by Nikita is
licensed under CC-BY 2.0

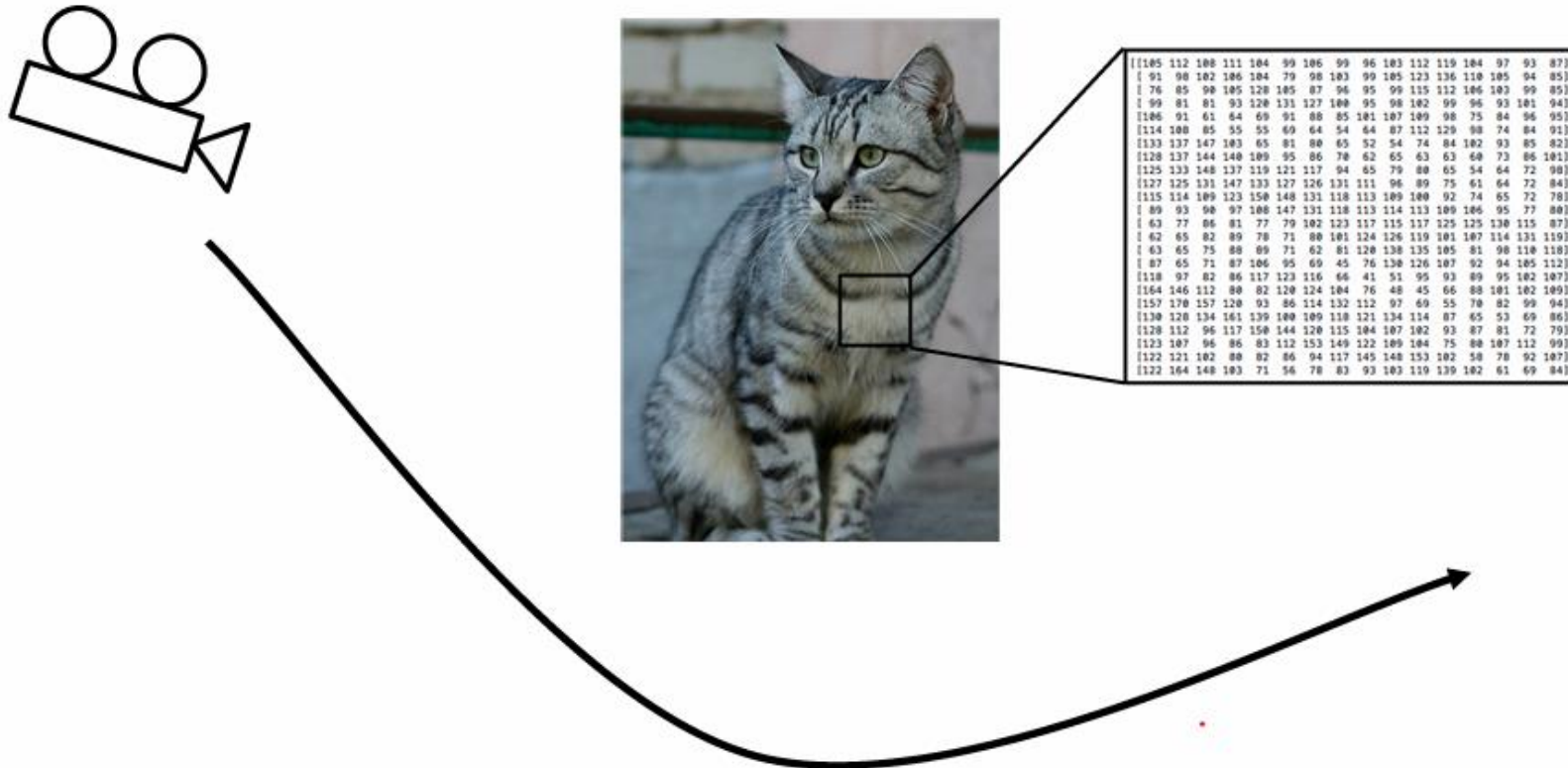
Output: Assign image to one
of a fixed set of categories

cat
bird
deer
dog
truck

$$f_{\theta}(x) = \{c_1, c_2, \dots, c_m\}$$

Challenges: Image Classification

- Viewpoint variation



All pixels change when
the camera moves!

Challenges: Image Classification

- Intraclass variation



Initial model

M_1 (gray)
cats
dogs
deer

M_2 (no stripes)
(orange)
dogs

Challenges: Image Classification

- Fine-grained categories

Fine-grain visual recognition

Maine Coon



Ragdoll



American Shorthair



Challenges: Image Classification

- Background clutter

noise



actual
object

↓
1



↑
2

Challenges: Image Classification

- Illumination changes
- Occlusion →
- Etc. etc.

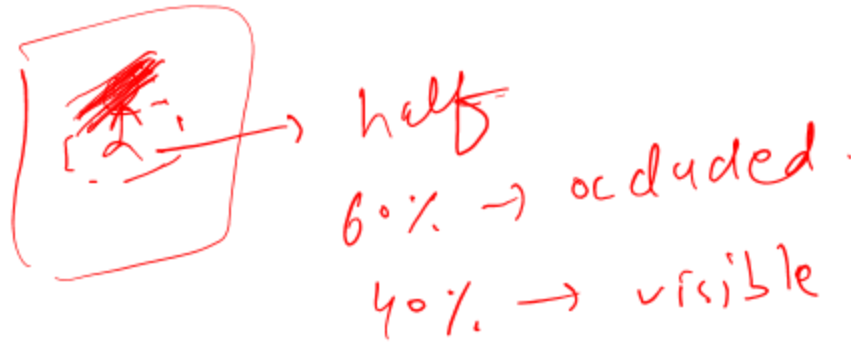


Image Classification

- Is image classification only useful for class assignment?
- Can we extend it? Can we use it somewhere else?

Image Classification

- Is image classification only useful for class assignment?
- Can we extend it? Can we use it somewhere else?
- **Is classification only useful for assigning a single class to an entire image? Or can we extend it further?**



→ Person
 → beach
 → horse
 → sky

} classify
 each
 object

Image Classification

- Is image classification only useful for class assignment?
- Can we extend it? Can we use it somewhere else?
- **We can use image classification as building block for other tasks such as object detection.**

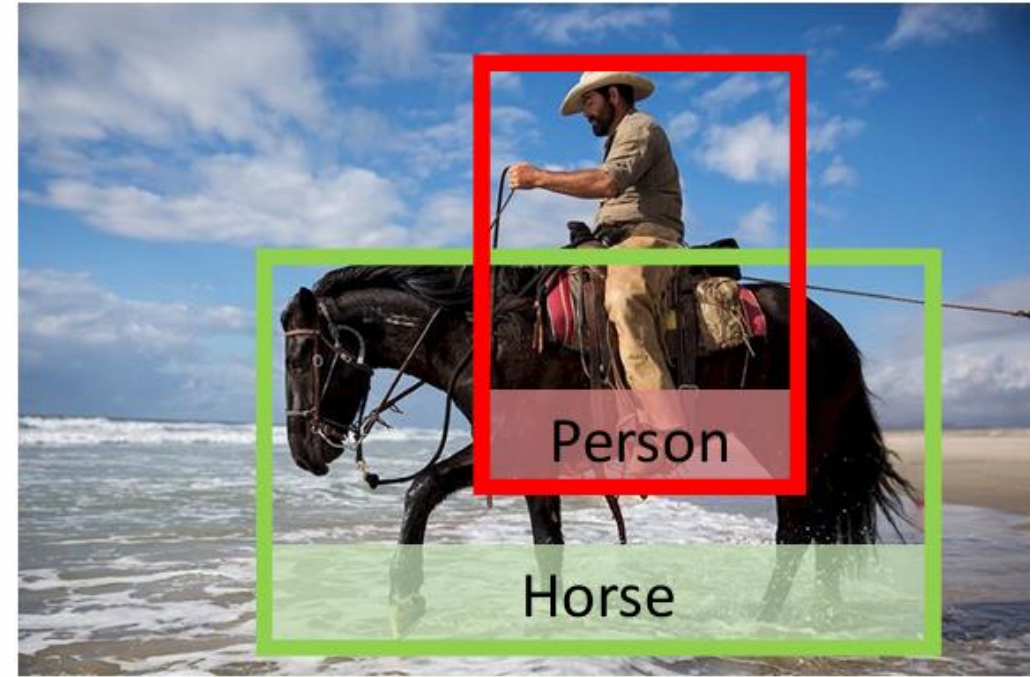


Image Classification

- Is image classification only useful for class assignment?
- Can we extend it? Can we use it somewhere else?
- **We can use image classification as building block for other tasks such as object detection.**

Image captioning -

✓ QA -



Background
Horse
Person
Car
Truck



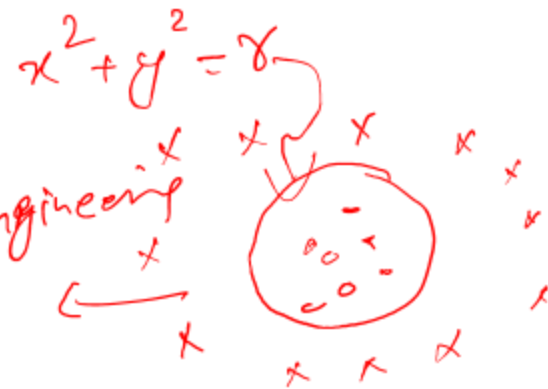
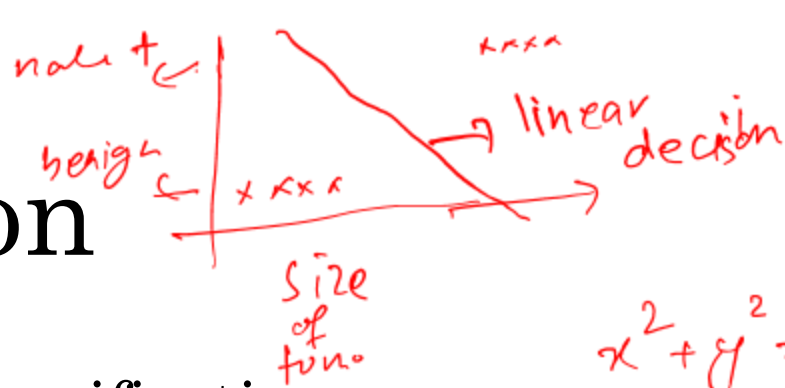
Background ✓
Horse ✓
Person ✓
Car
Truck

→ No of learnable parameters
→ No of operations.

Image Classification

classification.

- Logistic Regression for image classification.



Non-linear decision → feature engineering.



$$1000 \times 1000 = 10^6$$

$$\hat{y} = \sigma \left(\theta_0 + \sum_{i=1}^n \theta_i x_i \right) + \sum_{j=1}^6 \sum_{k=1}^6 \theta_{jk} x_{jk}$$

$$|\theta| = 10^6 + 10 \cdot 10^6$$

$$|\theta| \approx 10^{12} \rightarrow \text{greater} \geq \text{GPT4}$$



Image Classification

- Logistic Regression for image classification.
- Problem** > Curse of dimensionality.

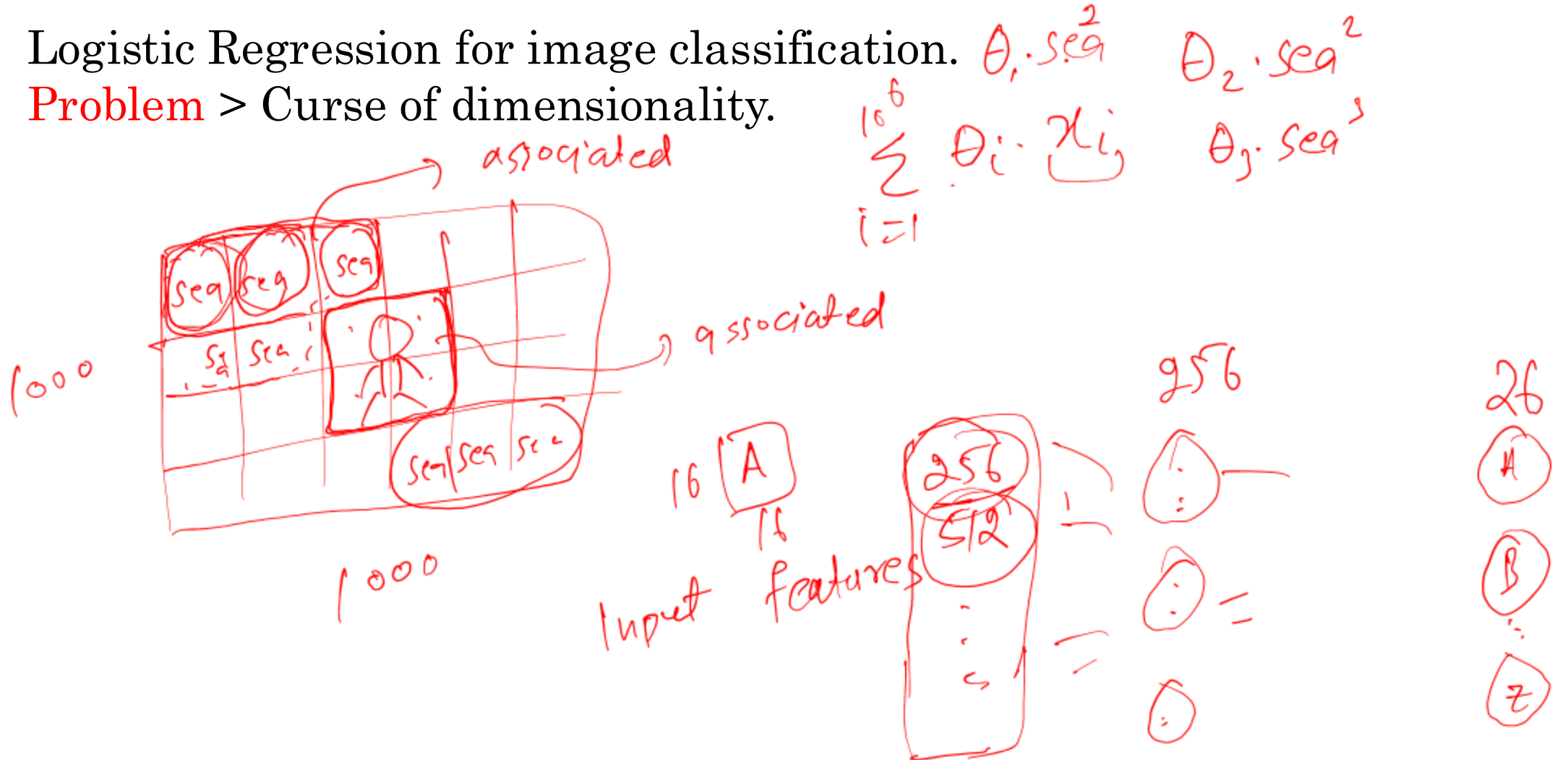
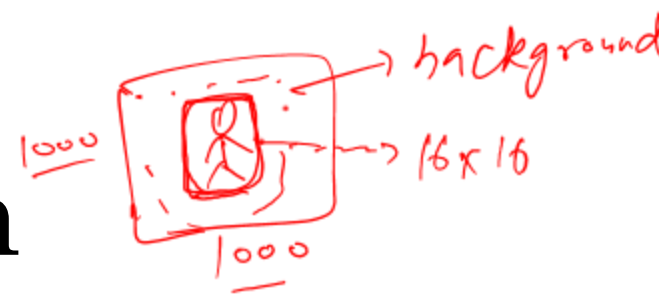


Image Classification



- Logistic Regression for image classification.
- **Problem:** Curse of dimensionality.
- **Solution:** Neural networks, select optimal features.

10^6 → inputs (pixels)

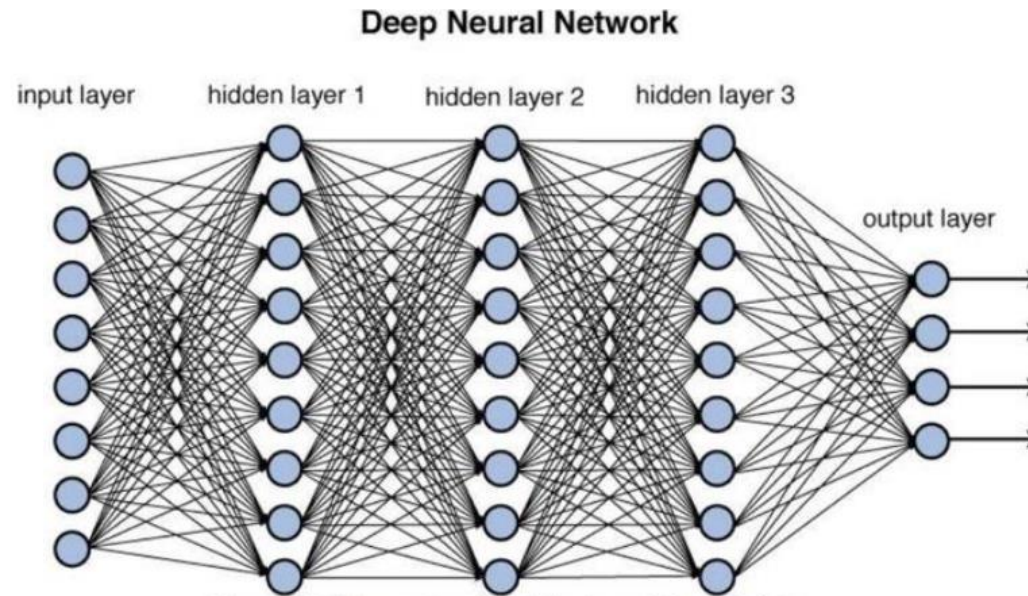


Figure 12.2 Deep network architecture with multiple layers.

10^6 → NN → 16×16
Input
learn $16 \times 16 / 32 \times 32$
most important features
for classification.

→ optimal features.

$$z = W_1x_1 + W_2x_2 + \dots + W_nx_n + b$$

$$\sum \delta(L_i)$$

Image Classification

- Neural networks for image classification

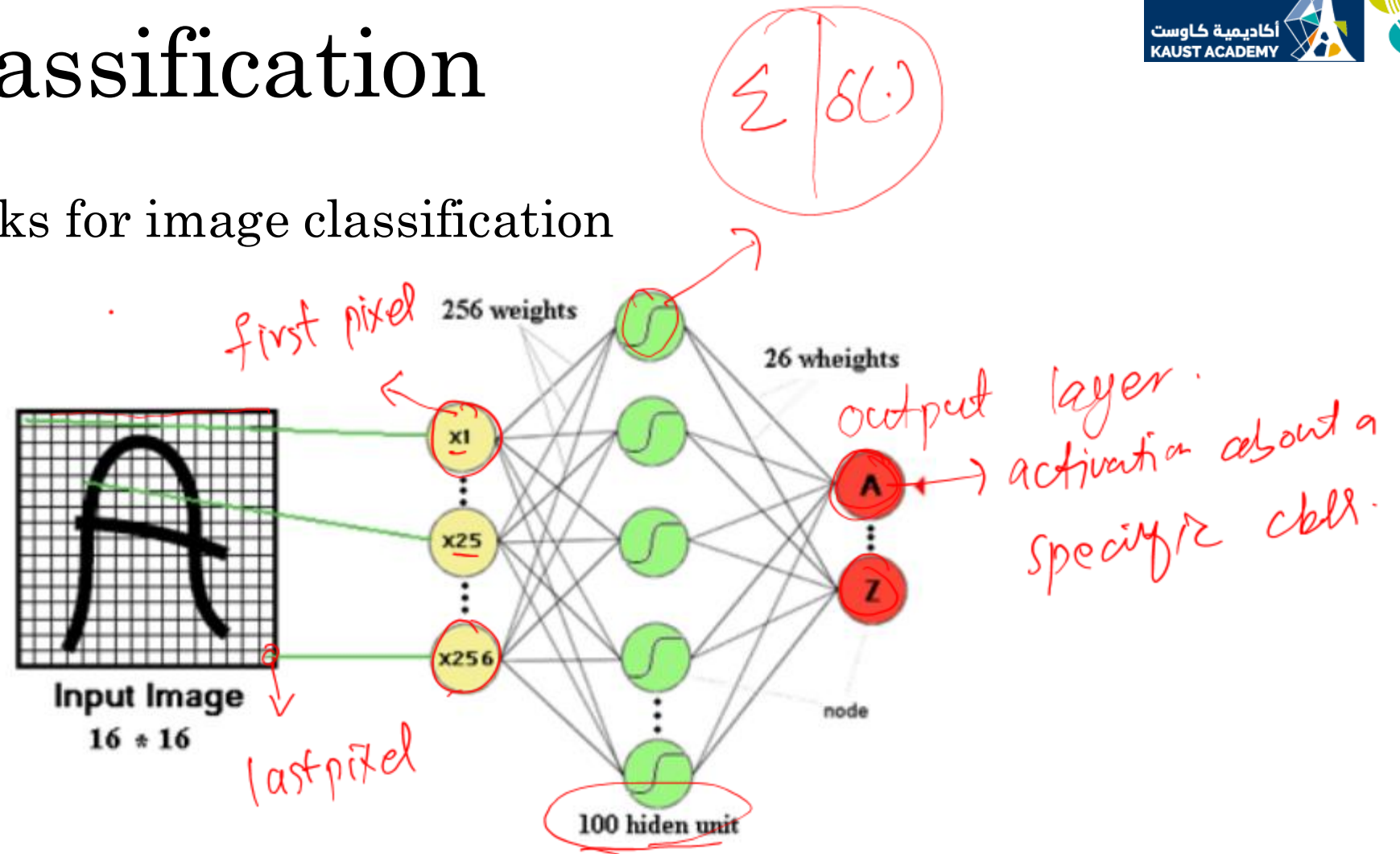
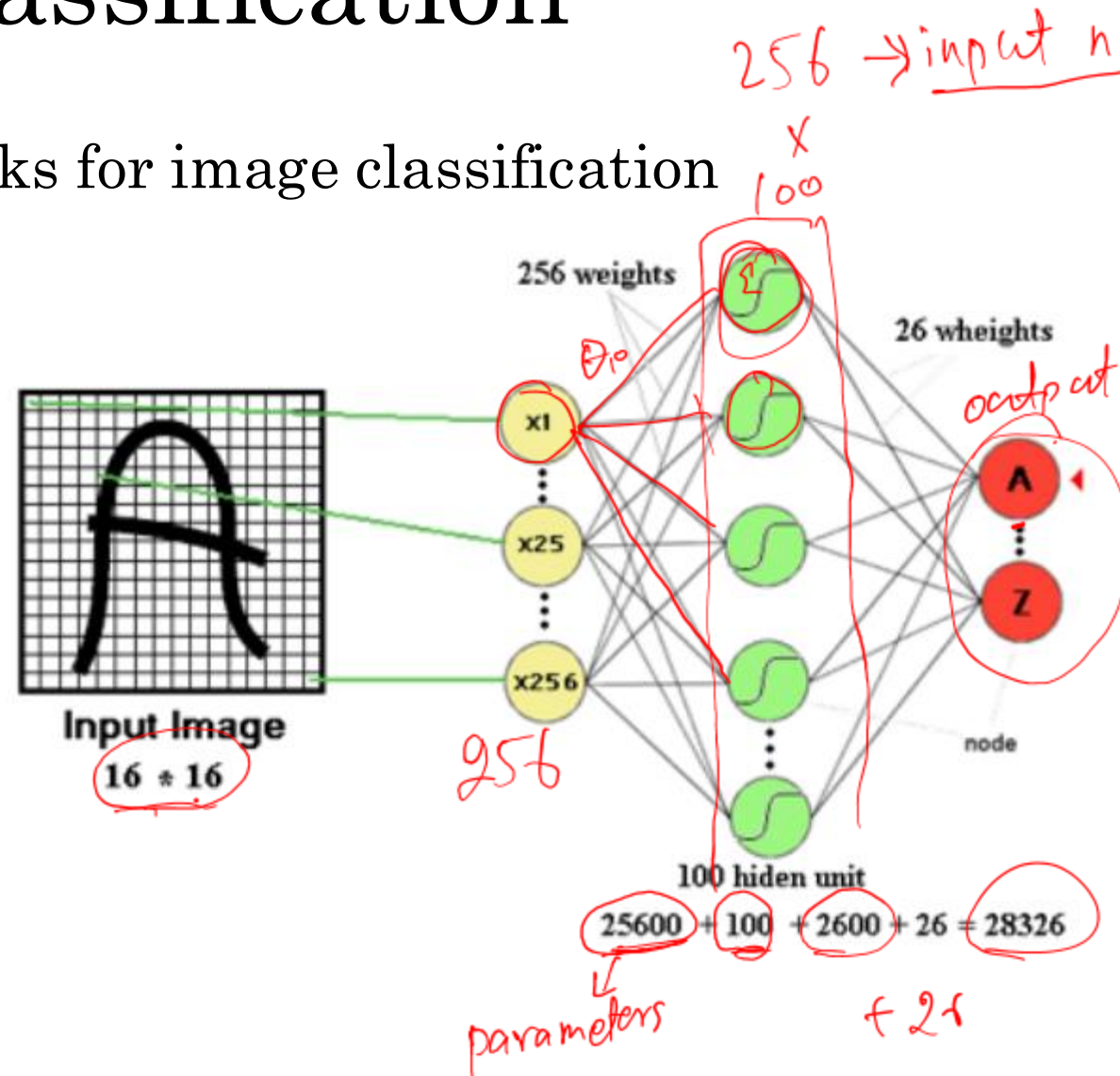


Image Classification

- Neural networks for image classification



$$\begin{matrix} \theta_{10} & \theta_{20} & \dots \\ \theta_{11} & \theta_{21} & \\ \theta_{12} & \theta_{22} & \\ \vdots & \vdots & \\ \theta_{1,100} & \theta_{2,100} & \end{matrix}$$

Image Classification

- Neural networks for image classification



64x64x3

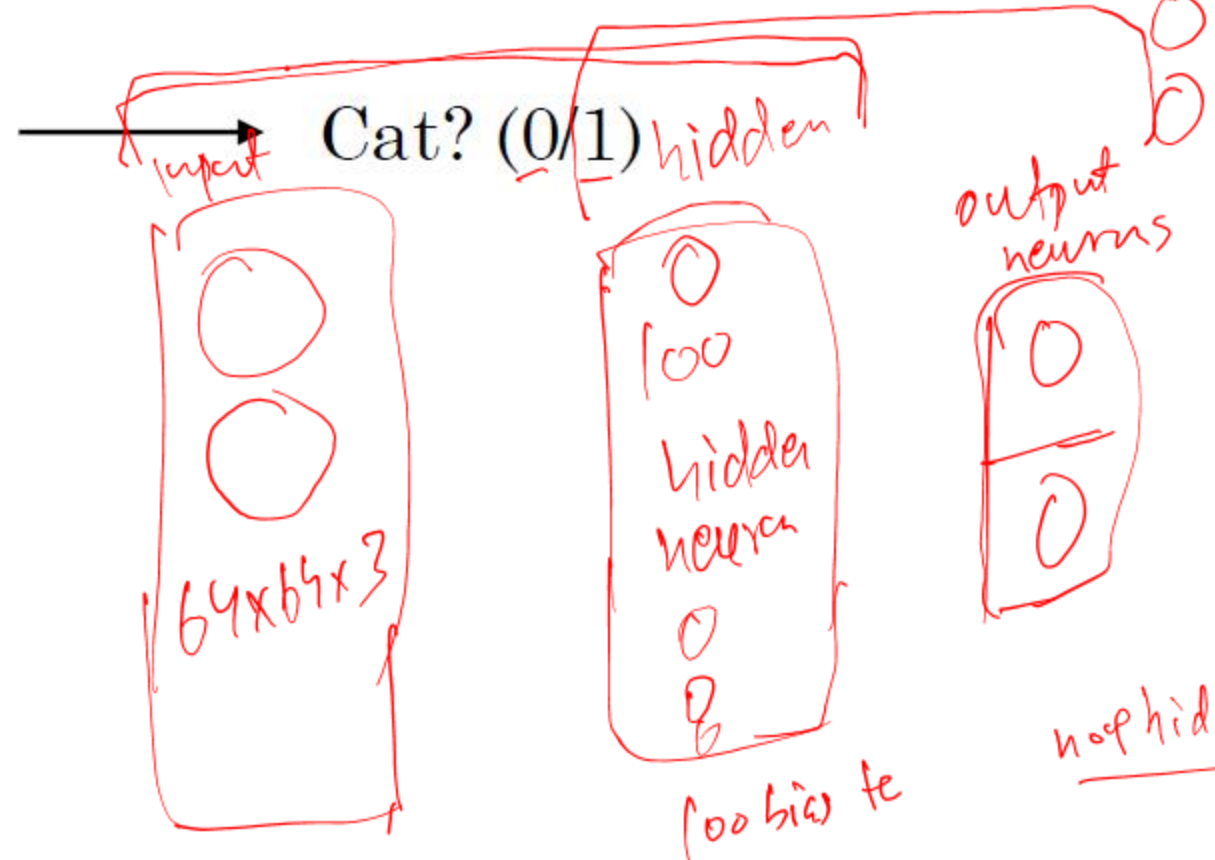


Image Classification

- Neural networks for image classification

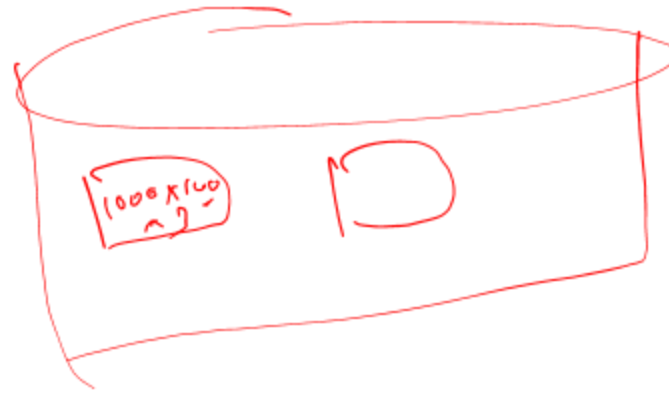


64x64x3

→ Cat? (0/1)



1000x1000x3



input neuron

6
10 x 3

pixels

3 hidden layers

100
200

500
1000

256
16x16

128
64
32

1st
2nd
3rd
4th

output (9)

10 1 3
2 4 8 1000

Image Classification

- Neural networks for image classification
- **Problem 1:** The number of parameters become extremely large, causing computation problem and overfitting.

References

- https://web.eecs.umich.edu/~justincj/slides/eecs498/WI2022/598_WI2022_lecture02.pdf
- https://web.eecs.umich.edu/~justincj/slides/eecs498/WI2022/598_WI2022_lecture03.pdf
- <https://towardsdatascience.com/training-deep-neural-networks-9fdb1964b964>